

7. 有 Ω 和 $\mathbb{R}$, 定义 $f: F \subseteq S \times \mathbb{R}$.

为高公理模态逻辑模态公理模态逻辑, 可以如此构造: $\mathcal{F} = \{ (w, r) \in \mathbb{Q} \times \mathbb{R} \mid \varphi(w, r) \}$

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Chapter 18 Random Variables

Similarly, M is a function mapping each outcome another way:

$$\begin{array}{ll} M(HHH) = 1 & M(THH) = 0 \\ M(HHT) = 0 & M(THT) = 0 \\ M(HTH) = 0 & M(TTH) = 0 \\ M(HTT) = 0 & M(TTT) = 1 \end{array}$$

So C and M are random variables.

18.1.1 Indicator Random Variables

An *indicator random variable* is a random variable that maps every outcome to either 0 or 1. Indicator random variables are also called *Bernoulli variables*. The random variable M is an example. If all three coins match, then $M = 1$; otherwise, $M = 0$.

Indicator random variables are closely related to events. In particular, an indicator random variable partitions the sample space into those outcomes mapped to 1 and those outcomes mapped to 0. For example, the indicator M partitions the sample space into two blocks as follows:

$$\underbrace{HHH \ TTT}_{M=1} \quad \underbrace{HHT \ HTH \ HTT \ THH \ THT \ TTH}_{M=0}$$

In the same way, an event E partitions the sample space into those outcomes in E and those not in E . So E is naturally associated with an indicator random variable, I_E , where $I_E(\omega) = 1$ for outcomes $\omega \in E$ and $I_E(\omega) = 0$ for outcomes $\omega \notin E$. Thus, $M = I_E$ where E is the event that all three coins match.

18.1.2 Random Variables and Events

There is a strong relationship between events and more general random variables as well. A random variable that takes on several values partitions the sample space into several blocks. For example, C partitions the sample space as follows:

$$\underbrace{TTT}_{C=0} \quad \underbrace{\cancel{TTH} \quad \cancel{THT} \quad HTT}_{C=1} \quad \underbrace{TTH \quad HTH \quad HHT}_{C=2} \quad \underbrace{HHH}_{C=3}.$$

Each block is a subset of the sample space and is therefore an event. So the assertion that $C = 2$ defines the event

$$[C = 2] = \{THH, HTH, HHT\},$$

and this event has probability

$$\Pr[C = 2] = \Pr[THH] + \Pr[HTH] + \Pr[HHT] = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = 3/8.$$

2. 从集合 Σ 到 Σ -代数, 为了处理处理无穷和不可测的奇特集合, 我们引入 Σ -测度的右半域 $\mathcal{F}$, 其明确了哪些子集是可以分配概率的.

⇒ 本质仍是等面积块, 不过单位面积, 某些单位面积大的小块其实是很多个等面积块的累加。由于趋近 0 着起耳, 测度差不多, 但数量完全不一致, 由度量积不定义, 给予“概率密度”!

definition: 概率函数 Pr 的定义域是事件 (样本空间的子集):

$Pr[E] = \sum_{\omega \in E} Pr[\omega]$, 即任何事件的概率都是其中样本点概率的和.

$$\Rightarrow Pr[R=\tau] \stackrel{\text{def}}{=} \sum_{\omega: R(\omega)=\tau} Pr[\omega]$$

$\Rightarrow$ 断言 $R=\tau$ 本身即一个事件 $\Rightarrow$ 或者说, 任何关于随机变量取值的断言都定义了一个事件.

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$$[R=\tau] = \{\omega \in S \mid R(\omega)=\tau\}$$

18.2. Independence

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$$\Rightarrow Pr[R=\tau]$$

$$= Pr[\{\omega \mid R(\omega)=\tau\}]$$

Likewise $[M=1]$ is the event $\{TTT, HHH\}$ and has probability $1/4$.

More generally, any assertion about the values of random variables defines an event. For example, the assertion that $C \leq 1$ defines

$$= \sum_{\omega: R(\omega)=\tau} Pr[\omega]$$

$$[C \leq 1] = \{TTT, TTH, THT, HTT\},$$

$\downarrow$

and so $Pr[C \leq 1] = 1/2$.

一个随机变量的概率, 不是完全产生的, 而是从样本空间里按规则“筛”出来的. Another example is the assertion that $C \cdot M$ is an odd number. If you think about it for a minute, you'll realize that this is an obscure way of saying that all three coins came up heads, namely,

$$[C \cdot M \text{ is odd}] = \{HHH\}.$$

$\Rightarrow$ 概率在本质上一定是可数的, 即, 随机变量之所以能进行概率运算, 是因为它最终会落到可数样本空间可数的基本事件上.

18.2 Independence

$\Rightarrow$ 即使在不可数样本空间上, 现代概率论也是在类似“筛”的思路下, 将任意随机事件分解为一组可数的子集类, 然后对每个子集类进行概率求和.

The notion of independence carries over from events to random variables as well. Random variables R_1 and R_2 are independent iff for all x_1, x_2 , the two events

对每个子集类进行概率求和.

$$[R_1 = x_1] \text{ and } [R_2 = x_2]$$

$\Rightarrow$ 即当看到 $Pr[R=\tau]$ 时, are independent.

其是在计算, 属于同一类. For example, are C and M independent? Intuitively, the answer should be “no.” The number of heads, C , completely determines whether all three coins match; that is, whether $M = 1$. But, to verify this intuition, we must find some $x_1, x_2 \in \mathbb{R}$ such that:

的结果, 它们的原始概率

$$Pr[C = x_1 \text{ AND } M = x_2] \neq Pr[C = x_1] \cdot Pr[M = x_2].$$

加起来是零.

One appropriate choice of values is $x_1 = 2$ and $x_2 = 1$. In this case, we have:

$$Pr[C = 2 \text{ AND } M = 1] = 0 \neq \frac{1}{4} \cdot \frac{3}{8} = Pr[M = 1] \cdot Pr[C = 2].$$

The first probability is zero because we never have exactly two heads ($C = 2$) when all three coins match ($M = 1$). The other two probabilities were computed earlier.

On the other hand, let H_1 be the indicator variable for the event that the first flip is a Head, so

$$[H_1 = 1] = \{HHH, HTH, HHT, HTT\}.$$

$$i) \Pr(R_1 = x_1, R_2 = x_2, \dots) = \Pr(R_1 = x_1) \dots \Pr(R_n = x_n).$$

$$\Rightarrow \Pr(R_1 = x_1 | R_2 = x_2) = \Pr(R_1 = x_1),$$

$$\text{即 } \Pr(A|B) = \Pr(A)$$

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Then H_1 is independent of M , since

$$\Pr[M = 1] = 1/4 = \Pr[M = 1 | H_1 = 1] = \Pr[M = 1 | H_1 = 0]$$

$$\Pr[M = 0] = 3/4 = \Pr[M = 0 | H_1 = 1] = \Pr[M = 0 | H_1 = 0]$$

This example is an instance of:

Lemma 18.2.1. *Two events are independent iff their indicator variables are independent.*

The simple proof is left to Problem 18.1.

Intuitively, the independence of two random variables means that knowing some information about one variable doesn't provide any information about the other one. We can formalize what "some information" about a variable R is by defining it to be the value of some quantity that depends on R . This intuitive property of independence then simply means that functions of independent variables are also independent:

Lemma 18.2.2. *Let R and S be independent random variables, and f and g be functions such that $\text{domain}(f) = \text{codomain}(R)$ and $\text{domain}(g) = \text{codomain}(S)$. Then $f(R)$ and $g(S)$ are independent random variables.*

The proof is another simple exercise left to Problem 18.30.

As with events, the notion of independence generalizes to more than two random variables.

Definition 18.2.3. Random variables $R_1, R_2, \dots, R_n$ are mutually independent iff for all $x_1, x_2, \dots, x_n$, the n events

$$[R_1 = x_1], [R_2 = x_2], \dots, [R_n = x_n]$$

are mutually independent. They are *k-way independent* iff every subset of k of them are mutually independent.

Lemmas 18.2.1 and 18.2.2 both extend straightforwardly to k -way independent variables.

18.3 Distribution Functions 分布函数.

A random variable maps outcomes to values. The probability density function, $\text{PDF}_R(x)$, of a random variable, R , measures the probability that R takes the value

x , and the closely related cumulative distribution function, $\text{CDF}_R(x)$, measures the probability that $R \leq x$. Random variables that show up for different spaces of outcomes often wind up behaving in much the same way because they have the same probability of taking different values, that is, because they have the same pdf/cdf.

Definition 18.3.1. Let R be a random variable with codomain V . The *probability density function* of R is a function $\text{PDF}_R : V \rightarrow [0, 1]$ defined by:

$$\text{PDF}_R(x) ::= \begin{cases} \Pr[R = x] & \text{if } x \in \text{range}(R), \\ 0 & \text{if } x \notin \text{range}(R). \end{cases}$$

If the codomain is a subset of the real numbers, then the *cumulative distribution function* is the function $\text{CDF}_R : \mathbb{R} \rightarrow [0, 1]$ defined by:

$$\text{CDF}_R(x) ::= \Pr[R \leq x].$$

A consequence of this definition is that

$$\sum_{x \in \text{range}(R)} \text{PDF}_R(x) = 1.$$

This is because R has a value for each outcome, so summing the probabilities over all outcomes is the same as summing over the probabilities of each value in the range of R .

As an example, suppose that you roll two unbiased, independent, 6-sided dice. Let T be the random variable that equals the sum of the two rolls. This random variable takes on values in the set $V = \{2, 3, \dots, 12\}$. A plot of the probability density function for T is shown in Figure 18.1. The lump in the middle indicates that sums close to 7 are the most likely. The total area of all the rectangles is 1 since the dice must take on exactly one of the sums in $V = \{2, 3, \dots, 12\}$.

The cumulative distribution function for T is shown in Figure 18.2: The height of the i th bar in the cumulative distribution function is equal to the *sum* of the heights of the leftmost i bars in the probability density function. This follows from the definitions of pdf and cdf:

$$\text{CDF}_R(x) = \Pr[R \leq x] = \sum_{y \leq x} \Pr[R = y] = \sum_{y \leq x} \text{PDF}_R(y).$$

It also follows from the definition that

$$\lim_{x \rightarrow \infty} \text{CDF}_R(x) = 1 \text{ and } \lim_{x \rightarrow -\infty} \text{CDF}_R(x) = 0.$$

21] 即 $f(x) = \Pr[R=x]$

⇒ 将随机变量取每个值的概率
写成一个函数。

definition: 1. PDF (Probability Density Function), 概率密度函数.

2. CDF (Cumulative Distribution Function), 累积分布函数.

$PDF_R(r) = Pr[R=r]$, R 恰好等于某个值的概率.

$CDF_R(r) = Pr[R \leq r]$, R 小于或等于某个值的概率.

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⇒ KE! 随机变量只是整理了一下样本空间, 将它划分为不同的区块, 每个区块中的不同元素 (即区块中的结果) 在这个随机变量下都有相同的值.

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⇒ 1. 函数无区别: Both PDF_R and CDF_R capture the same information about R , so take your choice. The key point here is that neither the probability density function nor the cumulative distribution function involves the sample space of an experiment.

$R: S \rightarrow T$, 都会根据函数相同这个等价关系, 将定义域 S 划分为互不相交的子集, 每个子集里的元素在 R 下具有相同的值. One of the really interesting things about density functions and distribution functions is that many random variables turn out to have the *same* pdf and cdf. In other words, even though R and S are different random variables on different probability spaces, it is often the case that

$$PDF_R = PDF_S.$$

In fact, some pdf's are so common that they are given special names. For example, the three most important distributions in computer science are the *Bernoulli distribution*, the *uniform distribution*, and the *binomial distribution*. We look more closely at these common distributions in the next several sections.

2. 划分对应等价关系

$$\omega_1 \sim \omega_2 \Leftrightarrow R(\omega_1) = R(\omega_2)$$

⇒ 它不只是一个函数, 更是一个

分类器.

即 KE

其用处在于逻辑抽象和降维计算.

eg. 1% 的故障率, 一车该换零件?

令 $R = \begin{cases} 1, & \text{故障} \\ 0, & \text{将故障} \end{cases}$

for some $p \in [0, 1]$. The corresponding cumulative distribution function is $F_p: \mathbb{R} \rightarrow [0, 1]$ where

$$f_p(0) = p, \text{ and}$$

$$f_p(1) = 1 - p,$$

$$F_p(x) ::= \begin{cases} 0 & \text{if } x < 0 \\ p & \text{if } 0 \leq x < 1 \\ 1 & \text{if } 1 \leq x. \end{cases}$$

$$E[R] = 1 \times 1\% + 0 \times (1 - 1\%)$$

⇒ $E[X] = E[R_1 + R_2 + R_3 + \dots]$ 18.3.2 Uniform Distributions 均匀分布.

$$= E[R_1] + E[R_2] + \dots$$

$$= 0.1\% + 0.1\% + \dots$$

$$= 1.$$

A random variable that takes on each possible value in its codomain with the same probability is said to be *uniform*. If the codomain V has n elements, then the *uniform distribution* has a pdf of the form

$$PDF \quad f: V \rightarrow [0, 1]$$

where

$$f(v) = \frac{1}{n}$$

⇒ 用于将逻辑化为算术, 将离散概率的局部.

for all $v \in V$.

⇒ 均匀的意思就是在给定的有限集合上, 没有哪个值比其他值更特殊, 更可能出现, 其和我们通常中“公平”的概念完全一致。

If the elements of V in increasing order are $a_1, a_2, \dots, a_n$, then the cumulative distribution function would be $F : \mathbb{R} \rightarrow [0, 1]$ where

$$\text{CDF } F(x) ::= \begin{cases} 0 & \text{if } x < a_1 \\ k/n & \text{if } a_k \leq x < a_{k+1} \text{ for } 1 \leq k < n \\ 1 & \text{if } a_n \leq x. \end{cases}$$

Uniform distributions come up all the time. For example, the number rolled on a fair die is uniform on the set $\{1, 2, \dots, 6\}$. An indicator variable is uniform when its pdf is $f_{1/2}$.

18.3.3 The Numbers Game

Enough definitions—let's play a game! We have two envelopes. Each contains an integer in the range $0, 1, \dots, 100$, and the numbers are distinct. To win the game, you must determine which envelope contains the larger number. To give you a fighting chance, we'll let you peek at the number in one envelope selected at random. Can you devise a strategy that gives you a better than 50% chance of winning?

For example, you could just pick an envelope at random and guess that it contains the larger number. But this strategy wins only 50% of the time. Your challenge is to do better.

So you might try to be more clever. Suppose you peek in one envelope and see the number 12. Since 12 is a small number, you might guess that the number in the other envelope is larger. But perhaps we've been tricky and put small numbers in *both* envelopes. Then your guess might not be so good!

An important point here is that the numbers in the envelopes may *not* be random. We're picking the numbers and we're choosing them in a way that we think will defeat your guessing strategy. We'll only use randomization to choose the numbers if that serves our purpose: making you lose!

Intuition Behind the Winning Strategy

People are surprised when they first learn that there is a strategy that wins more than 50% of the time, regardless of what numbers we put in the envelopes.

Suppose that you somehow knew a number x that was in between the numbers in the envelopes. Now you peek in one envelope and see a number. If it is bigger than x , then you know you're peeking at the higher number. If it is smaller than x , then you're peeking at the lower number. In other words, if you know a number x between the numbers in the envelopes, then you are certain to win the game.

The only flaw with this brilliant strategy is that you do *not* know such an x . This sounds like a dead end, but there's a cool way to salvage things: try to *guess* x !

☆☆ Question 我的问题是, 随机变量 X 为离散且以常性质量空间所有事件为可测子, 又能完备的保证所有事件均能以此种方式量度?

1. 函 R, S 限, 将 W 中每一个切谓元义进行替换, 而实数元义是主元的

2. 信息有损是必然的 (将复杂事件 ω 映射为 θ)，完备性的真諦在于“足够好”。

没有模型是绝对完美的, "mcs" — 2015/5/18 — 1:43 — page 739 — #747 只能根据经验来建立, 并能建立有效的理论, 我是“无有”的。

3. 如果第一阶段不移, 自然可以定义更多的随机变量, 其中由所有随机变量组成

8-我教, 我学, 我记, 我思, 我悟, 我创

4. 随机变量每样东西, 而
是可测函数 $\Rightarrow$ 对于轴上每个
我们感兴趣的事件, 其
概率与样空间中必须
是一个全体的事件 (即属于 σ -代数).

$\Rightarrow$ 即从任意点出发, 有 $\omega \rightarrow R \rightarrow R(\omega)$; 及值域有限, 日子集均有一个唯一的后继.

18.1 Random Variable Examples

Definition 18.1.1. A random variable R on a probability space is a total function whose domain is the sample space.

不事件都于样在全间中取 The codomain of R can be anything, but will usually be a subset of the real numbers. Notice that the name "random variable" is a misnomer; random variables are actually functions.

For example, suppose we toss three independent, unbiased coins. Let C be the number of heads that appear. Let $M = 1$ if the three coins come up all heads or all

$\Rightarrow$ 引申问题: R 和 M 产生, determines the values of C and M . For example, if we flip heads, tails, heads, then $C = 2$ and $M = 0$. If we flip tails, tails, tails, then $C = 0$ and $M = 1$. In effect, counts the number of heads, and M indicates whether all the coins match.

Since each outcome uniquely determines C and M , we can regard them as functions mapping outcomes to numbers. For this experiment, the sample space is:

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$.

而由一种确定的对应法则 Now C is a function that maps each outcome in the sample space to a number as

定义: τ 一个从 $\Sigma \rightarrow \mathbb{R}$ 的函数 follows:

τ 是一个规则, 给字母串赋值. 如:

给一个由 x 和 y 组成的串, $\tau(xy) = y$

$C(HHH) = 3$	$C(THH) = 2$
$C(HHT) = 2$	$C(THT) = 1$
$C(HTH) = 2$	$C(TTH) = 1$
$C(HTT) = 1$	$C(TTT) = 0$

这里的规则可以是一个显式公式, 可以是一个表格, 可以是一个算法或判定过程, 不需要用什么方法, 只要是个规则确定、有收义, 且对 Ω 中每一个 ω 都有一个值, 那它就是一个函数.

2. 而关于如何证明, 这是意义的一部分. 罗德里格说, $f: \Omega \rightarrow \mathbb{R}$ 的连续性, 并不要求我们就能“写出”或“计算”每个值, 其只要求我们有一个相对集合: $F \subseteq \Omega \times \mathbb{R}$.

满足: 1. 全性: $\forall x \in \Omega$, 都有 $y \in \mathbb{R}$, 使得 $(x, y) \in F$.

2° 传递性: $(x, y_1) \in F$ 且 $(x, y_2) \in F$, 则 $y_1 = y_2$. F 即为 $\sim$ 的图 (graph).

证明见翻頁 P₇₄₀ # 748.

Randomized Algorithms

The best strategy to win the numbers game is an example of a *randomized algorithm*—it uses random numbers to influence decisions. Protocols and algorithms that make use of random numbers are very important in computer science. There are many problems for which the best known solutions are based on a random number generator.

For example, the most commonly-used protocol for deciding when to send a broadcast on a shared bus or Ethernet is a randomized algorithm known as *exponential backoff*. One of the most commonly-used sorting algorithms used in practice, called *quicksort*, uses random numbers. You'll see many more examples if you take an algorithms course. In each case, randomness is used to improve the probability that the algorithm runs quickly or otherwise performs well.

18.3.4 Binomial Distributions

The third commonly-used distribution in computer science is the *binomial distribution*. The standard example of a random variable with a binomial distribution is the number of heads that come up in n independent flips of a coin. If the coin is fair, then the number of heads has an *unbiased binomial distribution*, specified by the pdf $f_n : [0..n] \rightarrow [0, 1]$:

$$f_n(k) ::= \binom{n}{k} 2^{-n}. \quad \cancel{f_n} = \cancel{f_n}$$

This is because there are $\binom{n}{k}$ sequences of n coin tosses with exactly k heads, and each such sequence has probability 2^{-n} .

A plot of $f_{20}(k)$ is shown in Figure 18.4. The most likely outcome is $k = 10$ heads, and the probability falls off rapidly for larger and smaller values of k . The falloff regions to the left and right of the main hump are called the *tails of the distribution*.

In many fields, including Computer Science, probability analyses come down to getting small bounds on the tails of the binomial distribution. In the context of a problem, this typically means that there is very small probability that something *bad* happens, which could be a server or communication link overloading or a randomized algorithm running for an exceptionally long time or producing the wrong result.

The tails do get small very fast. For example, the probability of flipping at most 25 heads in 100 tosses is less than 1 in 3,000,000. In fact, the tail of the distribution falls off so rapidly that the probability of flipping exactly 25 heads is nearly twice the probability of flipping exactly 24 heads *plus* the probability of flipping exactly

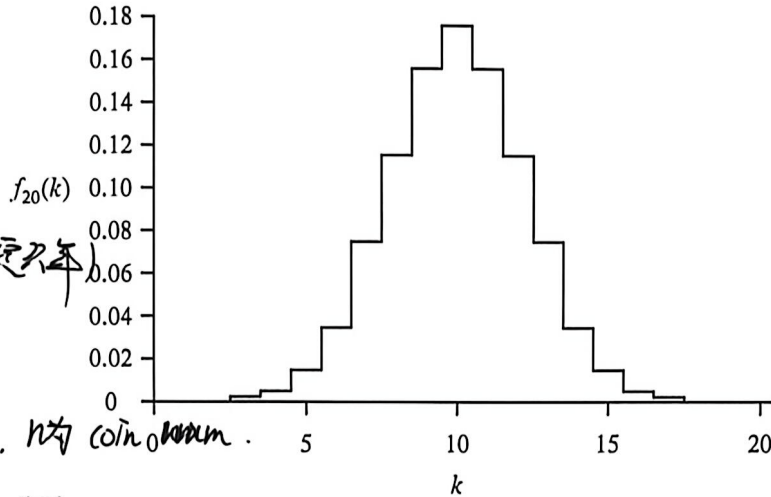


Figure 18.4 The pdf for the unbiased binomial distribution for $n = 20$, $f_{20}(k)$.

prob. 每个硬币 (并得一定年)
承而证明其独立.

⇒ 样本空间 2^n 个结果, n 为 coin num.

$$P = p^k \times q^{(n-k)}, \quad q = (1-p)$$

设 A B 分别为正, 负为反面.

$$Pr(A \cap B) = p \times p \times 1$$

⇒ 设有 1 正面, 剩下 $n-1$

$$Pr = \sum_{k=0}^{n-1} \binom{n-1}{k} p^k \times q^{(n-1-k)}$$

$$\Rightarrow (p+q)^{(n-1)}, \quad \text{而 } p+q=1$$

所以剩下来为 $\frac{1}{2}$

$$\text{即 } Pr(A \cap R) = Pr(A) Pr(R)$$

23 heads plus ... the probability of flipping no heads.

The General Binomial Distribution

If the coins are biased so that each coin is heads with probability p , then the number of heads has a *general binomial density function* specified by the pdf $f_{n,p}(k)$: $[0..n] \rightarrow [0, 1]$ where

$$f_{n,p}(k) = \binom{n}{k} p^k (1-p)^{n-k}. \quad \text{硬币 = 正.} \quad (18.1)$$

for some $n \in \mathbb{N}^+$ and $p \in [0, 1]$. This is because there are $\binom{n}{k}$ sequences with k heads and $n-k$ tails, but now $p^k (1-p)^{n-k}$ is the probability of each such sequence.

For example, the plot in Figure 18.5 shows the probability density function $f_{n,p}(k)$ corresponding to flipping $n = 20$ independent coins that are heads with probability $p = 0.75$. The graph shows that we are most likely to get $k = 15$ heads, as you might expect. Once again, the probability falls off quickly for larger and smaller values of k .

近似估计

Theorem 二项分布近似式

$$Pr(R=k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad \text{令 } k = \alpha n,$$

$$\binom{n}{\alpha n} \sim \frac{1}{\sqrt{2\pi n \alpha (1-\alpha)}} \cdot \alpha^{-\alpha n} \cdot (1-\alpha)^{-(1-\alpha)n}$$

$$\Rightarrow Pr(R=\alpha n) \sim \frac{\left[\left(\frac{p}{\alpha}\right)^\alpha \left(\frac{1-p}{1-\alpha}\right)^{1-\alpha} \right]^n}{\sqrt{2\pi n \alpha (1-\alpha)}}, \quad H(\alpha, p) = \alpha \log_2 \left(\frac{\alpha}{p}\right) + (1-\alpha) \log_2 \left(\frac{1-\alpha}{1-p}\right)$$

$$\Rightarrow Pr(R=\alpha n) \sim \frac{1}{\sqrt{2\pi n \alpha (1-\alpha)}} \cdot 2^{-n H(\alpha, p)}$$

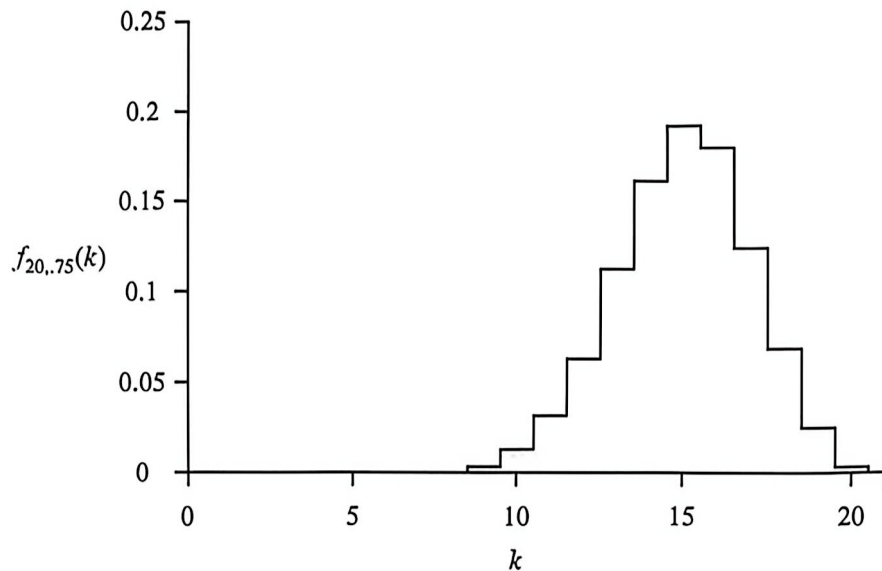


Figure 18.5 The pdf for the general binomial distribution $f_{n,p}(k)$ for $n = 20$ and $p = .75$.

18.4 Great Expectations

The *expectation* or *expected value* of a random variable is a single number that reveals a lot about the behavior of the variable. The expectation of a random variable is also known as its *mean* or *average*. For example, the first thing you typically want to know when you see your grade on an exam is the average score of the class. This average score turns out to be precisely the expectation of the random variable equal to the score of a random student.

More precisely, the expectation of a random variable is its “average” value when each value is weighted according to its probability. Formally, the expected value of a random variable is defined as follows:

Definition 18.4.1. If R is a random variable defined on a sample space S , then the expectation of R is

$$\text{Ex}[R] ::= \sum_{\omega \in S} R(\omega) \Pr[\omega]. \quad (18.2)$$

Let’s work through some examples.

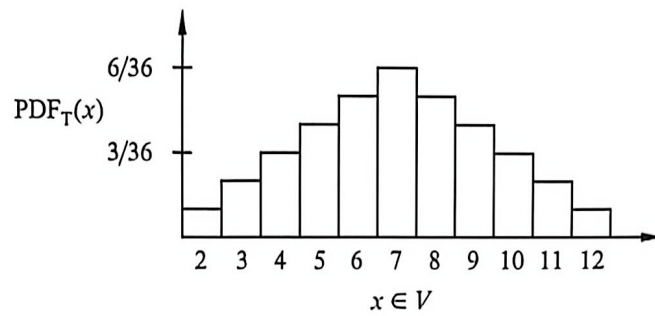


Figure 18.1 The probability density function for the sum of two 6-sided dice.

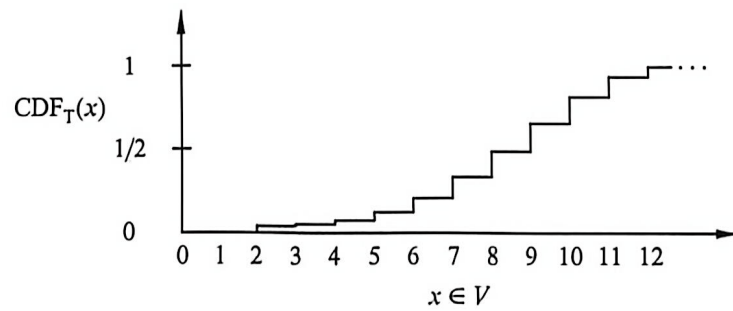


Figure 18.2 The cumulative distribution function for the sum of two 6-sided dice.

18.4.1 The Expected Value of a Uniform Random Variable

Rolling a 6-sided die provides an example of a uniform random variable. Let R be the value that comes up when you roll a fair 6-sided die. Then by (18.2), the expected value of R is

$$\text{Ex}[R] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = \frac{7}{2}.$$

This calculation shows that the name "expected" value is a little misleading; the random variable might *never* actually take on that value. No one expects to roll a $3\frac{1}{2}$ on an ordinary die!

In general, if R_n is a random variable with a uniform distribution on $\{a_1, a_2, \dots, a_n\}$, then the expectation of R_n is simply the average of the a_i 's:

$$\text{Ex}[R_n] = \frac{a_1 + a_2 + \dots + a_n}{n}.$$

18.4.2 The Expected Value of a Reciprocal Random Variable

Define a random variable S to be the reciprocal of the value that comes up when you roll a fair 6-sided die. That is, $S = 1/R$ where R is the value that you roll.

define median (中位数),

随机变量小于该值的概率

$$\text{Ex}[S] = \text{Ex}\left[\frac{1}{R}\right] = \frac{1}{1} \cdot \frac{1}{6} + \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{3} \cdot \frac{1}{6} + \frac{1}{4} \cdot \frac{1}{6} + \frac{1}{5} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} = \frac{49}{120}.$$

Notice that

$$\text{Ex}[1/R] \neq 1/\text{Ex}[R].$$

Assuming that these two quantities are equal is a common mistake.

随机变量大于该值的概率

等于 1 减去 小于 1/2

18.4.3 The Expected Value of an Indicator Random Variable

The expected value of an indicator random variable for an event is just the probability of that event.

Lemma 18.4.2. If I_A is the indicator random variable for event A , then

$$\text{Ex}[I_A] = \text{Pr}[A].$$

⇒ eg. 丢硬币.

X=3: $\text{Pr}(<3) = 3/6$ (V) Proof.

$$\text{Pr}(>3) = 1/2$$

$$\text{Ex}[I_A] = 1 \cdot \text{Pr}[I_A = 1] + 0 \cdot \text{Pr}[I_A = 0] = \text{Pr}[I_A = 1]$$

$$= \text{Pr}[A]. \quad (\text{def of } I_A)$$

✓ 4: $\text{Pr}(<4) = 1/2$ (V)

For example, if A is the event that a coin with bias p comes up heads, then $\text{Ex}[I_A] = \text{Pr}[I_A = 1] = p$.

$$\text{Pr}(>4) = 2/6$$

18.4.4 Alternate Definition of Expectation

There is another standard way to define expectation.

Theorem 18.4.3. *For any random variable R ,*

$$\text{Ex}[R] = \sum_{x \in \text{range}(R)} x \cdot \Pr[R = x]. \quad (18.3)$$

The proof of Theorem 18.4.3, like many of the elementary proofs about expectation in this chapter, follows by regrouping of terms in equation (18.2):

Proof. Suppose R is defined on a sample space $\mathcal{S}$. Then,

$$\begin{aligned} \text{Ex}[R] &::= \sum_{\omega \in \mathcal{S}} R(\omega) \Pr[\omega] \\ &= \sum_{x \in \text{range}(R)} \sum_{\omega \in [R=x]} R(\omega) \Pr[\omega] \\ &= \sum_{x \in \text{range}(R)} \sum_{\omega \in [R=x]} x \Pr[\omega] && (\text{def of the event } [R = x]) \\ &= \sum_{x \in \text{range}(R)} x \left(\sum_{\omega \in [R=x]} \Pr[\omega] \right) && (\text{factoring } x \text{ from the inner sum}) \\ &= \sum_{x \in \text{range}(R)} x \cdot \Pr[R = x]. && (\text{def of } \Pr[R = x]) \end{aligned}$$

The first equality follows because the events $[R = x]$ for $x \in \text{range}(R)$ partition the sample space $\mathcal{S}$, so summing over the outcomes in $[R = x]$ for $x \in \text{range}(R)$ is the same as summing over $\mathcal{S}$. ■

In general, equation (18.3) is more useful than the defining equation (18.2) for calculating expected values. It also has the advantage that it does not depend on the sample space, but only on the density function of the random variable. On the other hand, summing over all outcomes as in equation (18.2) sometimes yields easier proofs about general properties of expectation.

18.4.5 Conditional Expectation

Just like event probabilities, expectations can be conditioned on some event. Given a random variable R , the expected value of R conditioned on an event A is the probability-weighted average value of R over outcomes in A . More formally:

Definition 18.4.4. The *conditional expectation* $\text{Ex}[R \mid A]$ of a random variable R given event A is:

$$\text{Ex}[R \mid A] ::= \sum_{r \in \text{range}(R)} r \cdot \Pr[R = r \mid A]. \quad (18.4)$$

For example, we can compute the expected value of a roll of a fair die, given that the number rolled is at least 4. We do this by letting R be the outcome of a roll of the die. Then by equation (18.4),

$$\text{Ex}[R \mid R \geq 4] = \sum_{i=1}^6 i \cdot \Pr[R = i \mid R \geq 4] = 1 \cdot 0 + 2 \cdot 0 + 3 \cdot 0 + 4 \cdot \frac{1}{3} + 5 \cdot \frac{1}{3} + 6 \cdot \frac{1}{3} = 5.$$

Conditional expectation is useful in dividing complicated expectation calculations into simpler cases. We can find a desired expectation by calculating the conditional expectation in each simple case and averaging them, weighing each case by its probability.

For example, suppose that 49.6% of the people in the world are male and the rest female—which is more or less true. Also suppose the expected height of a randomly chosen male is 5' 11", while the expected height of a randomly chosen female is 5' 5." What is the expected height of a randomly chosen person? We can calculate this by averaging the heights of men and women. Namely, let H be the height (in feet) of a randomly chosen person, and let M be the event that the person is male and F the event that the person is female. Then

$$\begin{aligned} \text{Ex}[H] &= \text{Ex}[H \mid M] \Pr[M] + \text{Ex}[H \mid F] \Pr[F] \\ &= (5 + 11/12) \cdot 0.496 + (5 + 5/12) \cdot (1 - 0.496) \\ &= 5.6646 \dots \end{aligned}$$

which is a little less than 5' 8."

This method is justified by:

Theorem 18.4.5 (Law of Total Expectation). *Let R be a random variable on a sample space S , and suppose that $A_1, A_2, \dots$, is a partition of S . Then*

$$\text{Ex}[R] = \sum_i \text{Ex}[R \mid A_i] \Pr[A_i].$$

Proof.

$$\begin{aligned}
 \text{Ex}[R] &= \sum_{r \in \text{range}(R)} r \cdot \Pr[R = r] && \text{(by 18.3)} \\
 &= \sum_r r \cdot \sum_i \Pr[R = r \mid A_i] \Pr[A_i] && \text{(Law of Total Probability)} \\
 &= \sum_r \sum_i r \cdot \Pr[R = r \mid A_i] \Pr[A_i] && \text{(distribute constant } r) \\
 &= \sum_i \sum_r r \cdot \Pr[R = r \mid A_i] \Pr[A_i] && \text{(exchange order of summation)} \\
 &= \sum_i \Pr[A_i] \sum_r r \cdot \Pr[R = r \mid A_i] && \text{(factor constant } \Pr[A_i]) \\
 &= \sum_i \Pr[A_i] \text{Ex}[R \mid A_i]. && \text{(Def 18.4.4 of cond. expectation)}
 \end{aligned}$$

■

18.4.6 Mean Time to Failure 平均失效时间

A computer program crashes at the end of each hour of use with probability p , if it has not crashed already. What is the expected time until the program crashes? This will be easy to figure out using the Law of Total Expectation, Theorem 18.4.5. Specifically, we want to find $\text{Ex}[C]$ where C is the number of hours until the first crash. We'll do this by conditioning on whether or not the crash occurs in the first hour.

So define A to be the event that the system fails on the first step and $\bar{A}$ to be the complementary event that the system does not fail on the first step. Then the mean time to failure $\text{Ex}[C]$ is

$$\text{Ex}[C] = \text{Ex}[C \mid A] \Pr[A] + \text{Ex}[C \mid \bar{A}] \Pr[\bar{A}]. \quad (18.5)$$

Since A is the condition that the system crashes on the first step, we know that

$$\text{Ex}[C \mid A] = 1. \quad (18.6)$$

Since $\bar{A}$ is the condition that the system does *not* crash on the first step, conditioning on $\bar{A}$ is equivalent to taking a first step without failure and then starting over without conditioning. Hence,

$$\text{Ex}[C \mid \bar{A}] = 1 + \text{Ex}[C]. \quad (18.7)$$

Lecture 随机策略进阶策略

⇒ 为何确定性策略不可能成功? question

该确定性策略, 其问题在于 ⇒ 对手会知道你的策略.

即对手可以反向设计, 且任何固定规则均存在漏洞.

所以任何确定性策略在面对一个足够且对抗性的对手时, 都会被击败! KE!

"mcs" — 2015/5/18 — 1:43 — page 748 — #756

⇒ 关于随机性策略, 其利用了对方无从预判的性质.

748

Chapter 18 Random Variables

1. 即使随机切分点太密, 那只是退回均匀分布型, 没有损失.

2. 且随机性能让为“不可能被完美针对”.

⇒ 用博弈论来总结: 你的策略只是一个概率分布, 因此对手必须应对所有可能性, 从而平均下来得到一个正的期望收益.

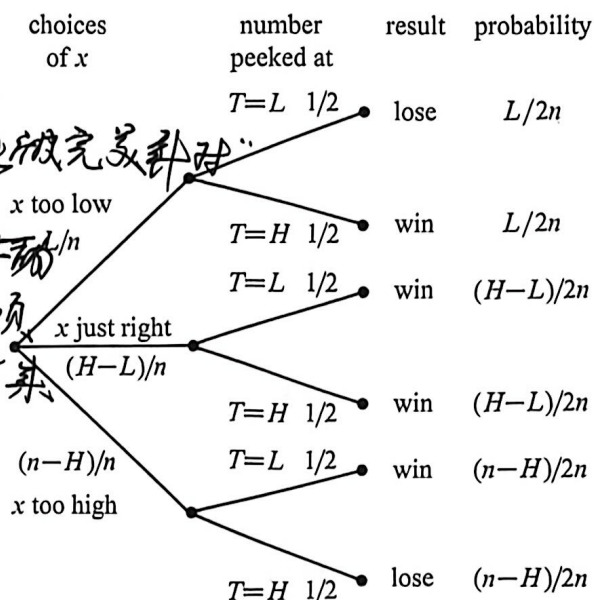


Figure 18.3 The tree diagram for the numbers game.

Step 4: Compute event probabilities.

The probability of the event that you win is the sum of the probabilities of the four outcomes in that event:

$$\begin{aligned} \Pr[\text{win}] &= \frac{L}{2n} + \frac{H-L}{2n} + \frac{H-L}{2n} + \frac{n-H}{2n} \\ &= \frac{1}{2} + \frac{H-L}{2n} \\ &\geq \frac{1}{2} + \frac{1}{2n} \end{aligned}$$

The final inequality relies on the fact that the higher number H is at least 1 greater than the lower number L since they are required to be distinct.

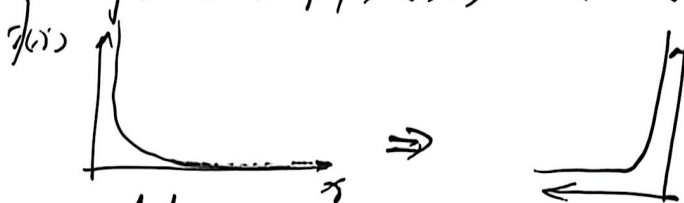
Sure enough, you win with this strategy more than half the time, regardless of the numbers in the envelopes! So with numbers chosen from the range $0, 1, \dots, 100$, you win with probability at least $1/2 + 1/200 = 50.5\%$. If instead we agree to stick to numbers $0, \dots, 10$, then your probability of winning rises to 55% . By Las Vegas standards, those are great odds.

lecture 数据分布有“长尾” (即中位数远超过平均值) 时, 其期望值可能非常大.

o.g. $\Pr(D \geq r) = 1/r$

$$\sum_{n=1}^{\infty} \frac{1}{n} \Rightarrow \int_1^{\infty} \frac{1}{x} dx = \ln(x) \Big|_1^{\infty} = \infty$$

"mcs" — 2015/5/18 — 1:43 — page 756 — #764



Plugging (18.6) and (18.7) into (18.5):

$$\begin{aligned} \text{Ex}[C] &= 1 \cdot p + (1 + \text{Ex}[C])(1 - p) \\ &= p + 1 - p + (1 - p) \text{Ex}[C] \\ &= 1 + (1 - p) \text{Ex}[C]. \end{aligned}$$

Can that problem

n个人的帽子打乱, 拿回自己帽子的期望

看初始变量 X_i : $\begin{cases} 1, & \text{if } i=1 \\ 0, & \text{if } i \neq 1 \end{cases}$

Then, rearranging terms gives

$$1 = \text{Ex}[C] - (1 - p) \text{Ex}[C] = p \text{Ex}[C],$$

and thus

$$\boxed{\text{Ex}[C] = 1/p.}$$

→ 每一步独立概率 p 的情况下发生

The general principle here is well-worth remembering.

Mean Time to Failure

If a system independently fails at each time step with probability p , then the expected number of steps up to the first failure is $1/p$.

So, for example, if there is a 1% chance that the program crashes at the end of each hour, then the expected time until the program crashes is $1/0.01 = 100$ hours.

As a further example, suppose a couple insists on having children until they get a boy, then how many baby girls should they expect before their first boy? Assume for simplicity that there is a 50% chance that a child will be a boy and that the genders of siblings are mutually independent.

This is really a variant of the previous problem. The question, "How many hours until the program crashes?" is mathematically the same as the question, "How many children must the couple have until they get a boy?" In this case, a crash corresponds to having a boy, so we should set $p = 1/2$. By the preceding analysis, the couple should expect a baby boy after having $1/p = 2$ children. Since the last of these will be a boy, they should expect just one girl. So even in societies where couples pursue this commitment to boys, the expected population will divide evenly between boys and girls.

There is a simple intuitive argument that explains the mean time to failure formula (18.8). Suppose the system is restarted after each failure. This makes the mean time to failure the same as the mean time between successive repeated failures. Now if the probability of failure at a given step is p , then after n steps we expect to have pn failures. Now, by definition, the average number of steps between failures is equal to np/p , namely, $1/p$.

转来问题: 即皮百佳
搞不明白

线性性 ~~→~~
↑

definition 主观概率是主观概率中“诚实”被试者真实报告的质盘空, 利率质量的
变所分配(不一定要和条件概率那样归一化)

贝叶斯 $Pr(A/B) = \frac{Pr(B/A) \cdot Pr(A)}{Pr(B)}$ Bress, Blackwell 以此建立决策论.

$\Rightarrow$ 相比较于量子方程, 概率论与其存在. 它的相似性

4. 列强 $\Rightarrow$ 概率果是全,

即随机变量之间的不独立.

即当确定了 x 的值, 则 $p(y/x)$ 会发生变化.

$\Rightarrow$ 端随机割是相同的. (条件 $p_i = \text{概率}$) $\Rightarrow$ 只有同一个线性微分方程.

P148 #756 关于以概率流动视角看随机策略博弈问题。

⇒ 本文基于以下6点泛泛给出：

1. Cenedin, A. (2016) Leven the Layer Number
2. Cover, T.M. (1987). Pick the Largest Number. In Cover, --
3. Bruss, F.T. (2003). Playing a trick on uncertainty
4. Hill, T.P. (2007). Knowing When to Stop.
5. Blackwell, D. (1951). On the Translating Parameter Problem for Discrete Van
6. Quanta Magazine. (2015, July 21). Solution: "Information from Randomness".

⇒ 概率流动模型

1° Static Probabilistic Mass, 静态概率质量。

definition 一个概率空间中，总质量 $Pr=1$ 守恒。在未引入任何条件前，所有可能结果共享一总质量。

⇒ 在 Quanta Magazine 中表述为 "a random guess is considered to have maximum disorder and consequently an information content of zero."

2° 概率锁死, Probabilistic Locking. KE!

definition 一个 Event (事件) 被观察到发生时，所有与之相关联的结果被排除，事件携带的概率被锁死，不再参与未来的概率分配，剩余质量可以保持，可以在随后的选择中重新下注并归一化。

其本质上即条件概率 $Pr(A|E) = \frac{Pr(A \cap E)}{Pr(E)}$ ，是条件概率的直观表现或成因。

3° Channel of Probabilistic Flow, 概率流动的通道

definition 一个通道是一个随机变量 T (独立于其他部分)，其选择性地将一部分概率质量从模糊状态引导至确定的状态，其本质特征是选择性渗透 ⇒ 只让满足特定条件的概率质量通过。

e.g. $Pr[Win] = \frac{1}{2} + Pr[Y < T \leq X]$ ，在拆塔时，概率会以一些情况流动；又如三门问题，开启一扇门后，该门的概率会流动至其他门。KE!

4° 随机化与通道问题

definition 随机化策略在暗中构建了一个由无数微小潜在通道构成的网络，为每轮游戏同时构建主部 ⇒ T 的累积分布函数以及必须维持界限（即支撑集无空隙）

Cenedin 2016 证明：若 $F \rightarrow G$ ，则 $\exists y < z$ ，Alice 恒有优势。

5° 贝叶斯更新作为条件流动。KE!

于是能够计算胜率, 得:

$$\frac{1}{2} p_k + \frac{1}{2} (1 - p_{k+1}) = \frac{1}{2} + \frac{1}{2} (p_k - p_{k+1})$$

Bob 与 $n-1$ 个相邻对上均匀随机序列, Alice 的期望胜率为

$$\frac{1}{n-1} \sum_{k=1}^{n-1} \left[\frac{1}{2} + \frac{1}{2} (p_k - p_{k+1}) \right] \quad \text{KE! 11}$$

而 $\sum_{k=1}^{n-1} (p_k - p_{k+1})$ 是一个 telescoping sum: $(p_1 - p_2) + (p_2 - p_3) \dots = p_1 - p_n$

因 $p_1, p_n \in [0, 1]$, 有 $p_1 - p_n \leq 1$, 于是 $E(W_m) = \frac{1}{2} + \frac{1}{2(n-1)} (p_1 - p_n)$

几乎给出了下界, 但 Bob 运用的 $n-1$ 个相邻对上, 得到的上界为 $\frac{1}{2} + \frac{1}{2(n-1)}$, 而 Alice 的期望 S_n 与 Bob 在总洗牌时为 $\frac{1}{2} + \frac{1}{2n}$.

⇒ 该问题来源于规范不统一, 将实数视为 $\{0, 1, \dots, m\}$, 其中 $m = n-1$, 则 Alice 为 $\frac{1}{2} + \frac{1}{2m}$, 与 Bob 的上界一致.

该结果已于 Gnedin (2016) — "Guess the Larger Number" (arXiv:1608.01899) 中得到了确认. $\square$

拓展: 假设连续, 有 $p(y < x = z) = F(z^-) - F(y^+) > 0$, 即连续分布具有支撑时, Alice 处于 $b(y, z)$ (非高斯分布) 上获得严格优势,

⇒ 随机化在 Alice 的策略中注入了独立性: 实值 x 的分布独立于 Bob 的数值时, 使得 Bob 无法通过而抵消 Alice 的优势.

11] 本处即取普适的期望公式: $\left[\frac{\sum p_i}{n} \right]$

论随机阈值策略为何优于纯猜测, 以及其为何最优. P_{48} #756
 $\Rightarrow$ 设有 $n \geq 2$ 个数, $\{1, 2, \dots, n\}$ 中秘密选取两个 $y < x$, 装入两个信封, Alice 从两个信封中随机打开一个, 看到 r , 并决定是保密还是竞猜. 其 Alice 目标是持有较大数 x .

★ **随机阈值策略**, 从策略 S_n 中 (半整数均匀阈值)

从 $T = \{\frac{1}{2}, 1\frac{1}{2}, 2\frac{1}{2}, \dots, n-\frac{1}{2}\}$ 中均匀地选取一个阈值 τ , 当 Alice 看到 r 时,
 1° 若 $r < \tau$, 则切换到另一个信封
 2° $r > \tau$, 则保留.
 注: $\tau \neq \tau$, τ 为半整数.

$\Rightarrow$ 胜率分析: (y, x) , 且 $y < x$.

Case 1. $y < \tau < x$, τ 为阈值.
 A 打开 y , $y < \tau$, 猜 $\Rightarrow x$ Win
 A 打开 x , $x > \tau$, 猜 $\Rightarrow x$ Win

Case 2. $\tau < y$ / $\tau > x$, 易思考得为等概率行为, 则 $\frac{1}{2}$ Lose, $\frac{1}{2}$ Win

$$\Rightarrow \Pr[\text{win} | (\tau < y) \vee (\tau > x)] = \frac{1}{2}$$

$\Rightarrow$ 其结果可由 tree 或: $\Pr[\text{win}] = \Pr[E] \cdot 1 + (1 - \Pr[E]) \cdot \frac{1}{2} = \frac{1}{2} + \frac{1}{2} \Pr[E]$.
 $E = \{y < \tau < x\} \Rightarrow$ 其概率大于 $\frac{1}{2}$, 其创造了正概率.

Worst Case 分析 (最坏下界)

$\Rightarrow$ T 的大小为 n , 每个阈值均有 $\frac{1}{n}$ 概率. τ 落入 (y, x) , 且 τ 严格介于 y 与 x 之间的概率为 $\frac{1}{n}$. 由于 y 与 x 之差 ≥ 1 , 则若 $x = y + 1$, 恰好一个. $\Rightarrow \Pr[E] = \frac{1}{n}$

$$\Pr[\text{Win}] \geq \frac{1}{2} + \frac{1}{2n}$$

$\Rightarrow$ **proof** S_n 为最优极大策略 $\Rightarrow$ 不存在 B 策略能在 Bob 的最优时优于策略 S_n 的胜率.

Theorem $\exists$ Alice 策略, Bob 可以迫使其胜率 $\leq \frac{1}{2} + \frac{1}{2n}$.

proof n . 若 Bob 采用如下的混合策略: 以 $\Pr_{\text{win}} = \frac{1}{n-1}$ 选择 $(k, k+1)$, $p \in \mathbb{R}$.

设 Alice 策略为, 看到 r , 以 p 猜, $(1-p)$ 保留, 则有:

1° 若概率猜 k : 猜, win / 不猜, lose.

2° $\frac{1}{2}$ 概率猜 $k+1$: 猜, lose / 不猜: Win

For the record, we'll state a formal version of this result. A random variable like C that counts steps to first failure is said to have a *geometric distribution* with parameter p .

Definition 18.4.6. A random variable, C , has a *geometric distribution* with parameter p iff $\text{codomain}(C) = \mathbb{Z}^+$ and

$$\Pr[C = i] = (1 - p)^{i-1} p.$$

Lemma 18.4.7. If a random variable C has a geometric distribution with parameter p , then

$$\text{Ex}[C] = \frac{1}{p}. \quad (18.8)$$

18.4.7 Expected Returns in Gambling Games

Some of the most interesting examples of expectation can be explained in terms of gambling games. For straightforward games where you win w dollars with probability p and you lose x dollars with probability $1 - p$, it is easy to compute your *expected return* or *winnings*. It is simply

$$pw - (1 - p)x \text{ dollars.}$$

For example, if you are flipping a fair coin and you win \$1 for heads and you lose \$1 for tails, then your expected winnings are

$$\frac{1}{2} \cdot 1 - \left(1 - \frac{1}{2}\right) \cdot 1 = 0.$$

In such cases, the game is said to be *fair* since your expected return is zero.

Splitting the Pot

We'll now look at a different game which is fair—but only on first analysis.

It's late on a Friday night in your neighborhood hangout when two new biker dudes, Eric and Nick, stroll over and propose a simple wager. Each player will put \$2 on the bar and secretly write "heads" or "tails" on their napkin. Then you will flip a fair coin. The \$6 on the bar will then be "split"—that is, be divided equally—among the players who correctly predicted the outcome of the coin toss. Pot splitting like this is a familiar feature in poker games, betting pools, and lotteries.

This sounds like a fair game, but after your regrettable encounter with strange dice (Section 16.3), you are definitely skeptical about gambling with bikers. So before agreeing to play, you go through the four-step method and write out the

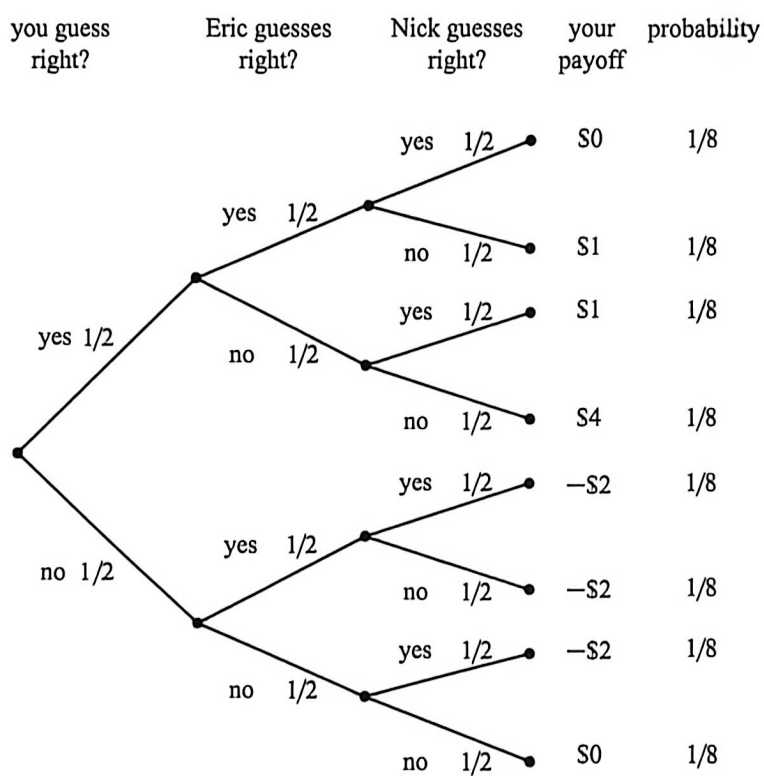


Figure 18.6 The tree diagram for the game where three players each wager \$2 and then guess the outcome of a fair coin toss. The winners split the pot.

tree diagram to compute your expected return. The tree diagram is shown in Figure 18.6.

The "payoff" values in Figure 18.6 are computed by dividing the \$6 pot¹ among those players who guessed correctly and then subtracting the \$2 that you put into the pot at the beginning. For example, if all three players guessed correctly, then your payoff is \$0, since you just get back your \$2 wager. If you and Nick guess correctly and Eric guessed wrong, then your payoff is

$$\frac{6}{2} - 2 = 1.$$

In the case that everyone is wrong, you all agree to split the pot and so, again, your payoff is zero.

To compute your expected return, you use equation (18.3):

$$\begin{aligned} \text{Ex}[\text{payoff}] &= 0 \cdot \frac{1}{8} + 1 \cdot \frac{1}{8} + 1 \cdot \frac{1}{8} + 4 \cdot \frac{1}{8} \\ &\quad + (-2) \cdot \frac{1}{8} + (-2) \cdot \frac{1}{8} + (-2) \cdot \frac{1}{8} + 0 \cdot \frac{1}{8} \\ &= 0. \end{aligned}$$

This confirms that the game is fair. So, for old time's sake, you break your solemn vow to never ever engage in strange gambling games.

The Impact of Collusion

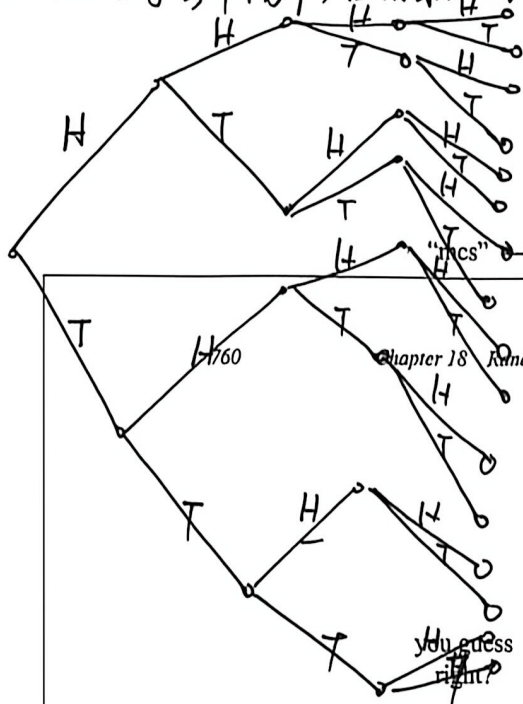
Needless to say, things are not turning out well for you. The more times you play the game, the more money you seem to be losing. After 1000 wagers, you have lost over \$500. As Nick and Eric are consoling you on your "bad luck," you do a back-of-the-envelope calculation and decide that the probability of losing \$500 in 1000 fair \$2 wagers is very, very small.

Now it is possible of course that you are very, very unlucky. But it is more likely that something fishy is going on. Somehow the tree diagram in Figure 18.6 is not a good model of the game.

The "something" that's fishy is the opportunity that Nick and Eric have to collude against you. The fact that the coin flip is fair certainly means that each of Nick and Eric can only guess the outcome of the coin toss with probability 1/2. But when you look back at the previous 1000 bets, you notice that Eric and Nick never made the same guess. In other words, Nick always guessed "tails" when Eric guessed "heads," and vice-versa. Modelling this fact now results in a slightly different tree diagram, as shown in Figure 18.7.

¹The money invested in a wager is commonly referred to as the *pot*.

① ② ③
 because 由于每个人互相猜对方硬币的解, 则有 $\Leftarrow$ 奈特策略.



1/10 均分

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若其中一人取对方的反 (不相同) 则得:
 即为样本空间, 且总概率的-入压(万) = -1/10.

而前问题, 解决/求解相类似

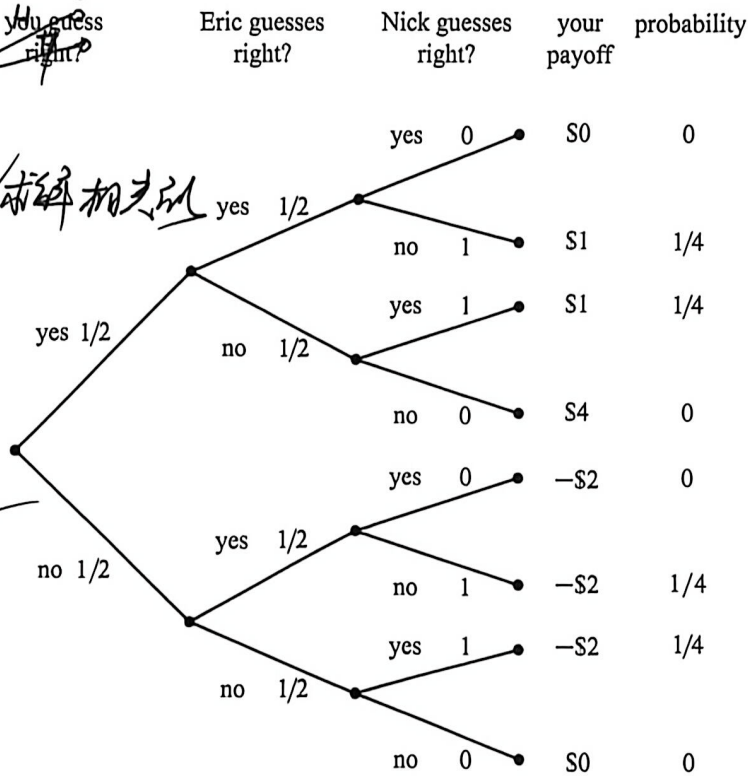


Figure 18.7 The revised tree diagram reflecting the scenario where Nick always guesses the opposite of Eric.

将硬币求成了...
 因求到规则与样对好.

期望值是博弈的产物，而非游戏规则固有的属性。

definition 反身性，即 chernoff 策略被破，并且导致所有参与者使用随机数，这导致将毁灭，游戏回到最初的“平等”状态。

在由具有将理性行为的群体构成的系统(市场)中，理性和随机性本身可以成为一种发展相对优势的策略。

18.4. Great Expectations

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The payoffs for each outcome are the same in Figures 18.6 and 18.7, but the probabilities of the outcomes are different. For example, it is no longer possible for all three players to guess correctly, since Nick and Eric are always guessing differently. More importantly, the outcome where your payoff is \$4 is also no longer possible. Since Nick and Eric are always guessing differently, one of them will always get a share of the pot. As you might imagine, this is not good for you!

When we use equation (18.3) to compute your expected return in the collusion scenario, we find that

$$\begin{aligned}\text{Ex}[\text{payoff}] &= 0 \cdot 0 + 1 \cdot \frac{1}{4} + 1 \cdot \frac{1}{4} + 4 \cdot 0 \\ &\quad + (-2) \cdot 0 + (-2) \cdot \frac{1}{4} + (-2) \cdot \frac{1}{4} + 0 \cdot 0 \\ &= -\frac{1}{2}.\end{aligned}$$

So watch out for these biker dudes! By colluding, Nick and Eric have made it so that you expect to lose \$0.50 every time you play. No wonder you lost \$500 over the course of 1000 wagers.

How to Win the Lottery

Similar opportunities to collude arise in many betting games. For example, consider the typical weekly football betting pool, where each participant wagers \$10 and the participants that pick the most games correctly split a large pot. The pool seems fair if you think of it as in Figure 18.6. But, in fact, if two or more players collude by guessing differently, they can get an "unfair" advantage at your expense!

In some cases, the collusion is inadvertent and you can profit from it. For example, many years ago, a former MIT Professor of Mathematics named Herman Chernoff figured out a way to make money by playing the state lottery. This was surprising since the state usually takes a large share of the wagers before paying the winners, and so the expected return from a lottery ticket is typically pretty poor. So how did Chernoff find a way to make money? It turned out to be easy!

In a typical state lottery,

- all players pay \$1 to play and select 4 numbers from 1 to 36,
- the state draws 4 numbers from 1 to 36 uniformly at random,
- the state divides 1/2 of the money collected among the people who guessed correctly and spends the other half redecorating the governor's residence.

This is a lot like the game you played with Nick and Eric, except that there are more players and more choices. Chernoff discovered that a small set of numbers

lecture 精哥某朝 supper court 的卡位。

由于规则为猜测得为不中奖，而该得为本身概率极不公平！

将 Page 761 # 769 中的 the 更改，基于一个平均为基础中所得，但公众本身也具猜以意。虽然好的在于得钱概率大，但人也多，年分下来钱很少，所以需将利率提高，即“最利下注点”

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只需找到该括号的真实发生概率远大于公众认为其发生的概率即可。

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Chapter 18 Random Variables

was selected by a large fraction of the population. Apparently many people think the same way; they pick the same numbers not on purpose as in the previous game with Nick and Eric, but based on the Red Sox winning average or today's date. The result is as though the players were intentionally colluding to lose. If any one of them guessed correctly, then they'd have to split the pot with many other players. By selecting numbers uniformly at random, Chernoff was unlikely to get one of these favored sequences. So if he won, he'd likely get the whole pot! By analyzing actual state lottery data, he determined that he could win an average of 7 cents on the dollar. In other words, his expected return was not $-\$0.50$ as you might think, but $+\$0.07$.² Inadvertent collusion often arises in betting pools and is a phenomenon that you can take advantage of.

★ 线性性不依赖于结构 ★

18.5 Linearity of Expectation

Expected values obey a simple, very helpful rule called *Linearity of Expectation*. Its simplest form says that the expected value of a sum of random variables is the sum of the expected values of the variables.

Theorem 18.5.1. For any random variables R_1 and R_2 ,

$$\text{Ex}[R_1 + R_2] = \text{Ex}[R_1] + \text{Ex}[R_2]$$

Proof. Let $T ::= R_1 + R_2$. The proof follows straightforwardly by rearranging terms in equation (18.2) in the definition of expectation:

$$\begin{aligned} \text{Ex}[T] &::= \sum_{\omega \in S} T(\omega) \cdot \text{Pr}[\omega] \\ &= \sum_{\omega \in S} (R_1(\omega) + R_2(\omega)) \cdot \text{Pr}[\omega] && (\text{def of } T) \\ &= \sum_{\omega \in S} R_1(\omega) \text{Pr}[\omega] + \sum_{\omega \in S} R_2(\omega) \text{Pr}[\omega] && (\text{rearranging terms}) \\ &= \text{Ex}[R_1] + \text{Ex}[R_2]. && (\text{by (18.2)}) \end{aligned}$$

A small extension of this proof, which we leave to the reader, implies

²Most lotteries now offer randomized tickets to help smooth out the distribution of selected sequences.

72] lecture and other implications

Linearity of Expectation
与独立性的关系：

期望线性方程并不适用于所有的独立情况：

72] 只有 $\sum_{i=0}^{\infty} \text{Ex}[R_i] < \infty$ 时才

成立，即：所有随机变量绝对值的期望之和必须有限。

$$\text{Ex}[\text{payoff}] = \sum (\text{Outcome} \times \text{Pr}(\text{Outcome}))$$

Lecture Tail Sum Formula, 尾数和公式.

$$E[R] = \sum_{i=0}^{\infty} P(R > i) \quad [7.1]$$

对于非负整数的随机变量 $[7.1] \Leftrightarrow E[R] = \sum_{k=0}^{\infty} k \cdot P(R=k) \Leftrightarrow E[R] = \sum_{i=0}^{\infty} P(R \geq i)$

易于证明, 只需展开即得 $\Rightarrow$ 以展开, 再以 k 为组, 变认于双重求和的重排.

"mcs" — 2015/5/18 — 1:43 — page 764 — #772

两者等价
(无穷求和)

1. 定义: 插值求和, Chapter 18 Random Variables

e.g. 插进 k 个时的概率

是 $\sum k \cdot P(R=k)$

his own hat back is $1/n$. There are many different probability distributions of hat permutations with this property, so we don't know enough about the distribution of G to calculate its expectation directly using equation (18.2) or (18.3). But linearity of expectation lets us sidestep this issue.

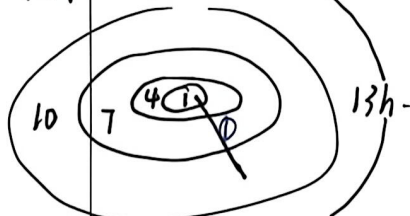
2. 尾数和: 尾数时程查

每时每刻时程查一次, 再累加

We'll use a standard, useful trick to apply linearity, namely, we'll express G as a sum of indicator variables. In particular, let G_i be an indicator for the event that the i th man gets his own hat. That is, $G_i = 1$ if the i th man gets his own hat, and $G_i = 0$ otherwise. The number of men that get their own hat is then the sum of these indicator random variables:

2] question 即此 $G = G_1 + G_2 + \dots + G_n$. (18.9)

来看: 设共 10 个可能



These indicator variables are *not* mutually independent. For example, if $n-1$ men all get their own hats, then the last man is certain to receive his own hat. But again, we don't need to worry about this dependence, since linearity holds regardless.

Since G_i is an indicator random variable, we know from Lemma 18.4.2 that

$$Ex[G_i] = Pr[G_i = 1] = 1/n. \quad (18.10)$$

By Linearity of Expectation and equation (18.9), this means that

$$\begin{aligned} Ex[G] &= Ex[G_1 + G_2 + \dots + G_n] \\ &= Ex[G_1] + Ex[G_2] + \dots + Ex[G_n] \\ &= \underbrace{\frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}}_n \\ &= 1. \end{aligned}$$

其中每一环均不平环, 即

4 & 7, 每个都为圆非环

现在我有 - 根线, 代表我等待

的时间: $0 = 1h + 4h + 7h + 10h$

这表时我有可能到达的时间

其概率为: $P_1 + P_4 + P_7 + P_{10}$

再分别乘上其大小, 最后得出

我真正可能的时间分布, 即求线.

概率由于均对其归一化, 所以对其分布无意义, 只对其自己有意义.

e.g. 对于 S_4 , 以知有 0.4 可能为 4h, 以有 0.6 可能为 1h, 7h, 10h 等, 以自身为归一化.

以 0 为可能考虑的区间, 那样本区间, 不同部分有不同“长度”, 即为权重 (待处理时间的可能性) $\Rightarrow$ 其长度即为权重

hat problem

P764 #772 期望本质

于柯莫哥拉夫公理体系 [Kolmogorov], 有一样基础: 样本空间 Ω , 其中所有可能结果的集合.

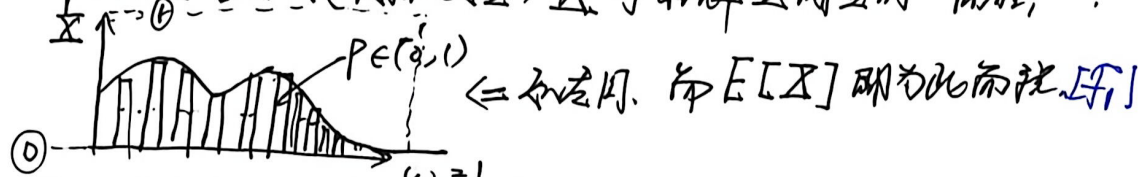
一个随机变量 X , 为一个函数: $X: \Omega \rightarrow \mathbb{R}$, 其为 $\forall \omega \in \Omega$ 举行量高度.

对于期望, 其测度论定义为 **面积**:

$\Rightarrow (\Omega, \mathcal{F}, P)$ 为一个概率空间, $X: \Omega \rightarrow \mathbb{R}$ 是一个随机变量. X 的 Expectation (期望) 记为 $E[X]$ 关于 P 的 Lebesgue 积分 (面积).

$$E[X] = \int_{\Omega} X(\omega) dP(\omega) \quad [7.1]$$

即: 期望是函数 (随机变量) X 于概率空间上的 "面积".



当 X 只有有限个值之间变化时 (离散随机变量), 该面积自然解释为所求/已知 "面积和".

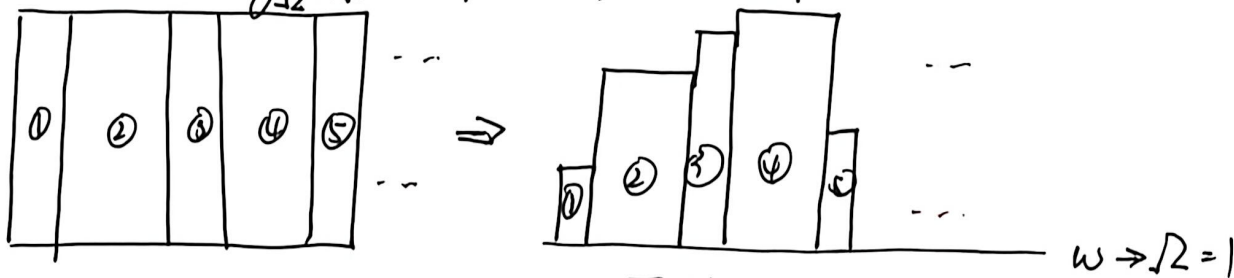
设 X 只有有限个值 $x_1, x_2, \dots, x_n$, 有 event: $A_i := \{\omega \in \Omega: X(\omega) = x_i\}$. 该集合 (n个) $A_1, \dots, A_n$, 构成了 Ω 的一个划分 (partition). $A_i \cap A_j = \emptyset$ ($i \neq j$), $A_1 \cup A_2 \cup \dots \cup A_n = \Omega$. **KE!** 于是有 $X(\omega) = \sum_{i=1}^n x_i \cdot 1_{A_i}(\omega)$ [7.2]

其中 $1_{A_i}(\omega) = 1$ 若 $\omega \in A_i$, 否则为 0. 于是有 Lebesgue integration

$$\begin{aligned} E[X] &= \int_{\Omega} X(\omega) dP(\omega) = \int_{\Omega} \sum_{i=1}^n x_i \cdot 1_{A_i}(\omega) dP(\omega) \\ &= \sum_{i=1}^n x_i \cdot \int_{\Omega} 1_{A_i}(\omega) dP(\omega) \end{aligned}$$

即示性函数的积分即为该集合的测度.

$$\int_{\Omega} 1_{A_i}(\omega) dP(\omega) = P(A_i), \text{ 即 } E(X) = \sum_{i=1}^n x_i \cdot P(A_i = x_i)$$



$$E(T) = \sum (\text{high} \times \text{width})$$

KE! 注: 期望并非只是分布, 而是质量分布 (PDF), 之间互不相交.

$\Rightarrow$ 见无限重迭过程的必然结局 $\Rightarrow$ 由大数定律与中心极限定理,

$\Rightarrow$ 有大数定律的证明, 才有了期望 "这个" 名字.

求和公式，从直觉冲受到测度论基础，P64

$E[R] = \sum_{k=1}^{\infty} P(R \geq k)$ 中， $\{R \geq 1\} \supset \{R \geq 2\} \supset \dots$ ，是否直接相加可行？

直觉源于双重计数。

$\Rightarrow \text{proof } P(R \geq k) = \sum_{i=k}^{\infty} P(R=i)$

$\Rightarrow \sum_{k=1}^{\infty} P(R \geq k) = \sum_{k=1}^{\infty} \sum_{i=k}^{\infty} P(R=i)$ ，由于所有项非负且小于1，则级数绝对收敛（可数）

则可以合法交换求和顺序：Tonelli于计数测度上的定理

$= \sum_{i=1}^{\infty} \sum_{k=1}^i P(R=i) = \sum_{i=1}^{\infty} i \cdot P(R=i) = E[R]$

	$k=1$ $R \geq 1$	$k=2$ $R \geq 2$	$k=3$ $R \geq 3$	$k=4$ $R \geq 4$	$k=5$ $R \geq 5$
$R=1$	✓	○	○	○	○
$R=2$	✓	✓	○	○	○
$R=3$	✓	✓	✓	○	○
$R=4$	✓	✓	✓	✓	○
$R=5$	✓	✓	✓	✓	✓

$\Rightarrow$ 按列相加时，为 $\sum_k P(R \geq k)$

$\Downarrow$
按行相加时，为 $\sum_i i \cdot P(R=i) = E[R]$ ，
 $i=(1, 2, \dots, t)$

$\Downarrow$
该论证于非负项下绝对合法，其严格性由 Tonelli 定理保证。

$\Rightarrow$ 该公式并非一个陌生的计数技巧，而是更深层数学原理的自然推论。

勒贝格积分有一个直观的“层蛋糕”表示 (Layer-Cake Representation)

$E[R] = \int_0^{\infty} P(R > t) dt$ ，离散下 $= \sum_{k=0}^{\infty} P(R \geq k+1) = \sum_{k=1}^{\infty} P(R \geq k)$

$\Rightarrow$ 其本质在于 Fubini 定理——对非负可测函数，二重平面上的面积与累测积分

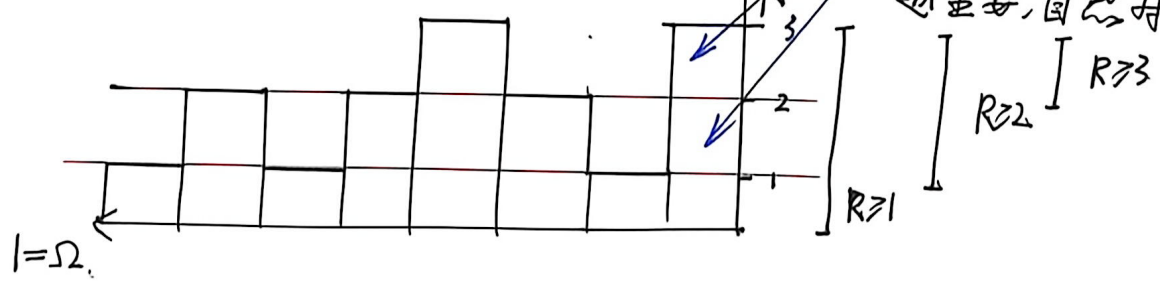
等价 (直观理解)，传统期望：问“爆炸的平均亮度” $\Rightarrow$ 将每个值乘上概率再相加

层蛋糕：1. 对于每一个高度等级，去测量其亮度时间。★

2. 将所有亮度等级的持续时间加起来

$\Downarrow$ 3. 更亮的事件，会被不断计数，权重相同。

R 越大，本身跟一种权重，若将 R 理解为值， R 越大， R 越重要，自然对平均亮度



附: 积分公式的证明.

$$\text{proof } E[X] = \int_0^{\infty} x f(x) dx$$

$$\text{令 } x = \int_0^x 1 dt = \int_0^{\infty} I_{\{t \leq x\}} dt$$

$$E[X] = \int_0^{\infty} x f(x) dx = \int_0^{\infty} \left(\int_0^{\infty} I_{\{t \leq x\}} dt \right) f(x) dx$$

$$\text{Fubini-Tonelli} \Rightarrow = \int_0^{\infty} \int_0^{\infty} I_{\{t \leq x\}} f(x) dx dt$$

$$= \int_0^{\infty} \left(\int_0^{\infty} I_{\{t \leq x\}} f(x) dx \right) dt$$

$$= \int_0^{\infty} \left(\int_t^{\infty} f(x) dx \right) dt$$

$$= \int_0^{\infty} P(X > t) dt$$

P166. Lecture

Thm. 2 $Pr(T \geq 1) \leq E[T]$

即为“至少发生一次”的概率 ≤ 一个随机变量期望值的上界。

e.g. - 一个系统有 100 个故障，若这些故障发生的期望值加起来，即为 1。那么，立刻可以推出：

“出事”的概率很小。

proof | 尾数和， $E[T] = \sum_{i=1}^{\infty} Pr[T \geq i] \geq Pr[T \geq 1]$ $\square$

代入过程 - 则有

col | $Pr[T \geq 1] \leq \sum_{i=1}^n Pr[A_i] \Rightarrow$ 若故障有依赖关系，直接加总。

但，e.g. 假设这 100 个故障是两两相互独立，一个发生其他也会发生呢？

若有 $\forall i, j: R(A_i/A_j) = 1$ ，即一个故障依赖所有

proof: $\frac{Pr(A_i \cap A_j)}{Pr(A_j)} = 1$ 且 $Pr(A_j \cap A_i) = Pr(A_i)$ (对称性)

而 $Pr(A_i \cap A_j) = Pr(A_j \cap A_i)$ ，则 $Pr(A_i) = Pr(A_j) = p [F]$

则 $Pr(A_i \cap A_j) = p$

不断递进，则所有的故障发生概率 p 为同一值。

显然，这样做的前提是所有故障发生的概率相同 $[F]$ ，与是否独立。

$\Rightarrow$ 于是引入了更般化且估计的 **Murphy's Law** 墨菲定律。

Thm. 3 $A_1 \sim A_n$ 相互独立， T 为发生事件数，有 $Pr[T=0] \leq e^{-E[T]}$

proof: $Pr(T=0) = \prod_{i=1}^n Pr(\bar{A}_i) = \prod_{i=1}^n (1 - Pr(A_i))$ ，又有 $1-x \leq e^{-x}$

则有 $\prod_{i=1}^n (1 - Pr(A_i)) \leq \prod_{i=1}^n e^{-Pr(A_i)} = e^{-\sum_{i=1}^n Pr(A_i)}$

即：如果一堆坏事彼此独立，并且为事件 $= e^{-E[T]}$ $\square$

我们在生活中发生的坏事多，那么“一件都不发生”的概率小的不可思议

e.g. 有“滚雪球”效应

Thm. 4 $R_1, \dots, R_n$ 相互独立，有 $E[R_1 \cdot R_2 \cdot \dots] = E[R_1] \cdot E[R_2] \cdot \dots$

且独立事件相关性， $E(R) \neq E(R)^2$

Proof. Define R_i to be the indicator random variable for A_i , where $R_i(\omega) = 1$ if $\omega \in A_i$ and $R_i(\omega) = 0$ if $\omega \notin A_i$. Let $R = R_1 + R_2 + \cdots + R_n$. Then

$$\begin{aligned} \text{Ex}[R] &= \sum_{i=1}^n \text{Ex}[R_i] && \text{(by Linearity of Expectation)} \\ &= \sum_{i=1}^n \Pr[R_i = 1] && \text{(by Lemma 18.4.2)} \\ &= \sum_{i=1}^n \Pr[A_i]. && \text{(def of indicator variable)} \end{aligned}$$

So whenever you are asked for the expected number of events that occur, all you have to do is sum the probabilities that each event occurs. Independence is not needed.

18.5.3 Expectation of a Binomial Distribution

Suppose that we independently flip n biased coins, each with probability p of coming up heads. What is the expected number of heads?

Let J be the random variable denoting the number of heads. Then J has a binomial distribution with parameters n , p , and

$$\Pr[J = k] = \binom{n}{k} p^k (1-p)^{n-k}.$$

Applying equation (18.3), this means that

$$\text{Ex}[J] = \sum_{k=0}^n k \Pr[J = k] = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k}. \quad (18.11)$$

This sum looks a tad nasty, but linearity of expectation leads to an easy derivation of a simple closed form. We just express J as a sum of indicator random variables, which is easy. Namely, let J_i be the indicator random variable for the i th coin coming up heads, that is,

$$J_i ::= \begin{cases} 1 & \text{if the } i \text{th coin is heads} \\ 0 & \text{if the } i \text{th coin is tails.} \end{cases}$$

Then the number of heads is simply

$$J = J_1 + J_2 + \cdots + J_n.$$